We have so far seen the need for a quantum description of light by studying the blackbody spectrum of an object at temperature T, and the photoelectric effect. In both cases, the phenomena observerved were explained if light were assumed to be composed of discrete quanta (photons) with energy $E = hf$. If light is composed of particles with energy, do those particles also carry momentum?

In Newtonian mechanics, an object of mass m moving at velocity $\vec{v}$ has (kinetic) energy $E = \frac{1}{2}mv^2$ and momentum $\vec{p} = m\vec{v}$. A photon moves at the speed of light $v = c$ but it has no mass $m = 0$, so it appears it should have no kinetic energy and no momentum. However, Newtonian mechanics governs objects that are travelling at speeds slow compared to the speed of light, so it cannot be valid for light itself.

Special relativity governs how things behave at or near the speed of light, and the relativistic relation for energy and momentum is

$$E^2 = p^2c^2 + m^2c^4$$

This gives the familiar rest mass energy $E = mc^2$ for an object of mass m sitting at rest. The energy E is the kinematic energy, which is the rest mass energy plus the kinetic energy. Notice that when $m = 0$, such as for photons, the relation gives:

$$E = pc \quad \Rightarrow \text{photons carry only kinetic energy but also momentum}$$

We know that a photon's energy is $E = hf$, so its momentum must be

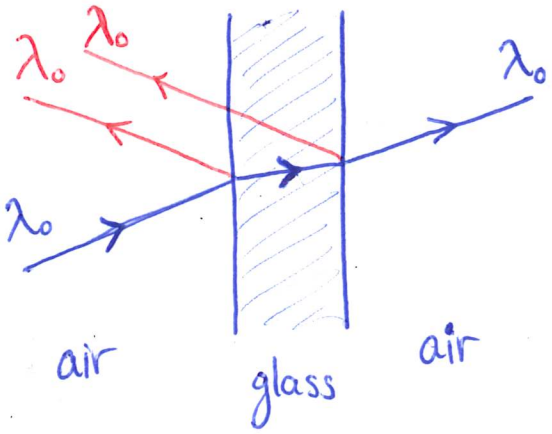
$$p = \frac{E}{c} = \frac{hf}{c} \qquad \text{but} \quad \lambda f = c$$

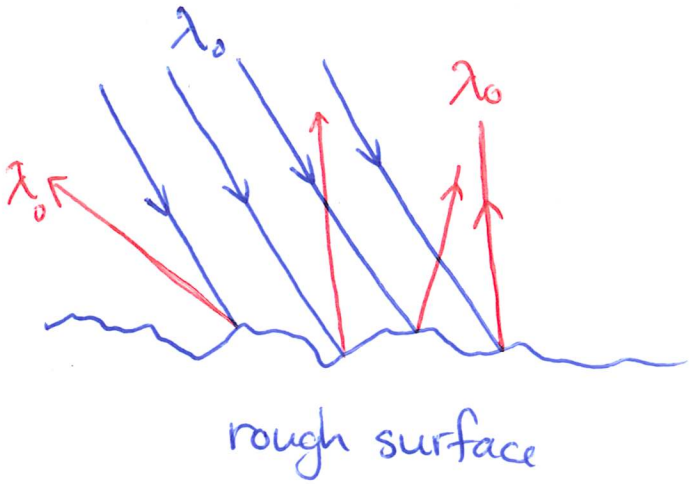
$$\boxed{p = \frac{h}{\lambda}}$$ photon's energy related to frequency f and its momentum is related to wavelength λ

Recall for any wave $\lambda = \frac{2\pi}{k}$ where k is the wavenumber

So we can re-express this as $p = \frac{hk}{2\pi}$. We can also define the reduced Planck's constant $\hbar \equiv \frac{h}{2\pi}$ to write this more succinctly as $\boxed{\vec{p} = \hbar\vec{k}}$ where $\vec{k}$ is called the wavevector and has magnitude $|\vec{k}| = k$ and its direction is aligned with the movement of the photon. Using $\hbar$ we can also equivalently express the energy $\boxed{E = \hbar\omega}$.

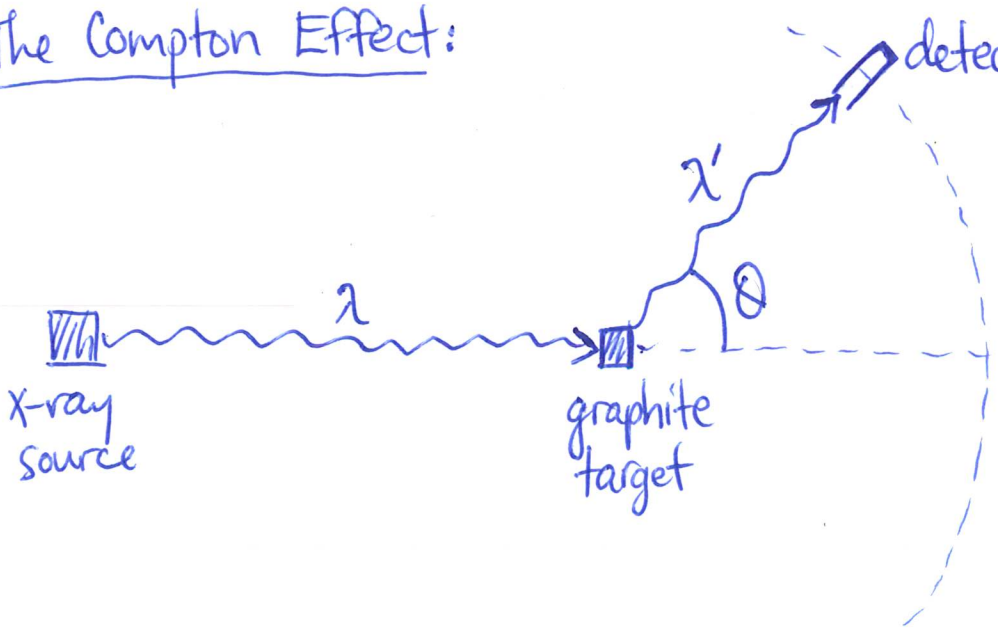
Why do we need to care about photon momentum? It is another prediction of quantum mechanics that helps explain how light interacts with matter. For example: from a purely classical perspective, when light interacts with a medium, the scattered light should have the same wavelength as the incident light.





However, when X-rays are scattered off certain materials like a graphite target, the scattered rays have different wavelengths than the incident rays, which has no explanation in classical physics treating the X-rays purely as a wave.

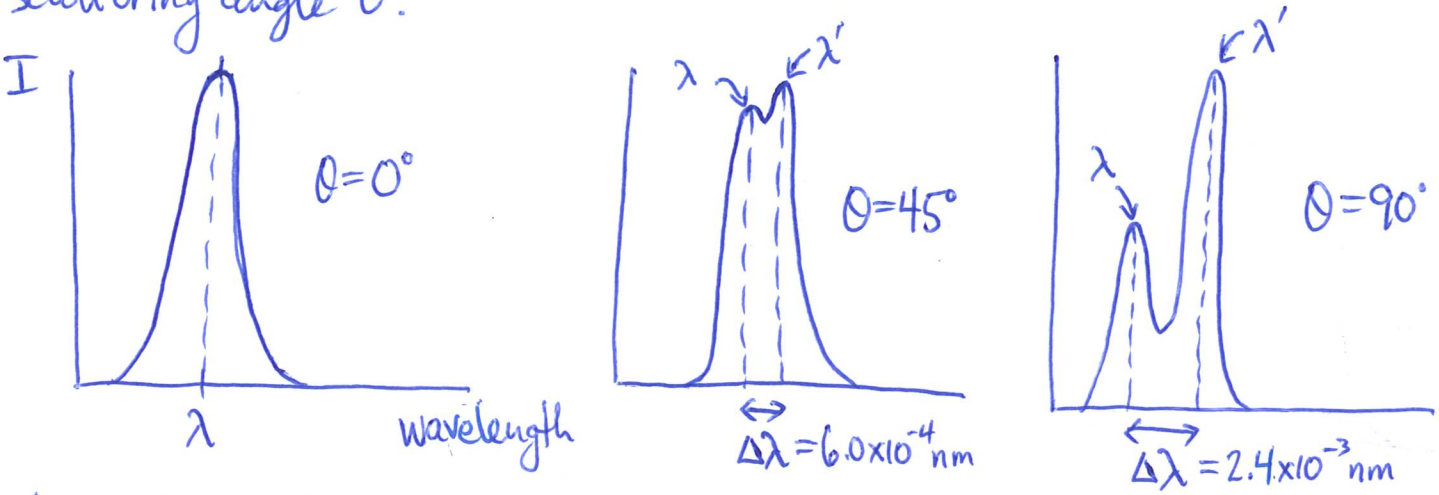
The Compton Effect:



X-rays of wavelength λ hit a graphite target and a detector sweeps through a range of deflection angles measuring the shifted wavelength λ' of the scattered rays.

5-minute break

What is observed in this Compton Scattering experiment at some scattering angle θ?



When $\theta=0$ there is no scattering and the detected $\lambda' = \lambda$. At non-zero scattering angle there is a second peak showing that some of the photons are having a non-classical interaction with the atoms in the graphite target.

Compton explained this in terms of the photons undergoing collisions with the free valence electrons in the graphite atoms: the incoming photon hits a valence electron and transfers some of its energy and momentum to it. The scattered photon thus carries away a lower energy (longer wavelength) than the incident photon. A full calculation of energy and momentum conservation yields the following:

$$\boxed{\Delta\lambda = \lambda_c(1-\cos\theta)}$$

where $\boxed{\lambda_c = \frac{h}{m_e c}}$ is the Compton wavelength of the electron.

$\lambda_c = 2.43\times10^{-12}\,m$

Example: An incident X-ray source has wavelength $\lambda = 71.0\,\text{pm}$ and is incident on a calcite target. What is the shifted wavelength at a scattering angle of 30°? What is it for 60°?

Solution: We know $\Delta\lambda = \lambda_c(1-\cos\theta)$ where $\lambda_c = \frac{h}{m_e c} = 2.43\,\text{pm}$

At $\theta = 30°$: $\Delta\lambda = (2.43\,\text{pm})\left(1 - \frac{\sqrt{3}}{2}\right) = 0.326\,\text{pm}$

therefore $\lambda' = \lambda + \Delta\lambda = 71.0\,\text{pm} + 0.3\,\text{pm} = \boxed{71.3\,\text{pm}}$

At $\theta = 60°$ $\Delta\lambda = (2.43\,\text{pm})\left(1 - \frac{1}{2}\right) = 1.215\,\text{pm}$

therefore $\lambda' = \lambda + \Delta\lambda = 71.0\,\text{pm} + 1.2\,\text{pm} = \boxed{72.2\,\text{pm}}$

Compton scattering works at any scattering angle, so what is the angle that yields the maximum shift?

$\Delta\lambda = \lambda_c(1-\cos\theta) \Rightarrow$ want to extremize $\Delta\lambda$

$\frac{d}{d\theta}\Delta\lambda = 0 \Rightarrow 0 = \lambda_c\left[-(-\sin\theta)\right]$

2 solutions:

① $\theta = 0$ (no shift)

② $\theta = 180°$ (max shift)

the maximum shift is thus at $\theta = 180°$ (i.e. back-scattered)

$\Rightarrow \boxed{\Delta\lambda_{max} = 2\lambda_c}$